

The empirical recurrence for OEIS A237725, recovered from its tail

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Abstract

OEIS A237725 records “Empirical recurrence of order 88” and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 1510$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 4$, so the recurrence is proved for every $n \geq 92$. Its last coefficient is $c_{88} = -41281126400 \neq 0$, so no further tail satisfies anything shorter; any order-88 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry’s recurrence is the one displayed here, whatever its linked file contains, and it is proved.

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1 A conjecture that is not written down

OEIS A237725 is “Number of $(n+1) \times (3+1)$ 0..2 arrays with the maximum plus the upper median minus the lower median of every 2×2 subblock differing from its horizontal and vertical neighbors by exactly one.”. It has offset 1 and begins

1510, 13496, 122940, 1145290, 10499035, 95855164, 880137130, 8127250116, ...

The entry states:

Empirical recurrence of order 88 (see link above)

As of the “Last modified” line on the live entry (Jul 23 2025 10:23:21, revision 7) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry’s condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u , and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) = \iota^\top M^{j(n)} \tau$$

for a linear reindexing j fixed by the entry’s shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 1510$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 1510$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \dots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S' + 1$ consecutive values of a . $\square$

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 88: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 4$ is the first whose minimal order is exactly the stated 88.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 3020$ exact terms from $n = 4$ on, it returns order 88 — the order the entry states — with integer coefficients:

$$a(n) = \sum_{i=1}^{88} c_i a(n-i),$$

c_1	0	c_{23}	0	c_{45}	0	c_{67}	0
c_2	0	c_{24}	−761001883797850677	c_{46}	−11148747851872695857784	c_{68}	−3083473706
c_3	0	c_{25}	0	c_{47}	0	c_{69}	0
c_4	9276	c_{26}	−431908363731757189	c_{48}	−7630215736422251635503	c_{70}	−337906690
c_5	0	c_{27}	0	c_{49}	0	c_{71}	0
c_6	118160	c_{28}	46356429818243542846	c_{50}	−1979054879561788228676	c_{72}	3622533460
c_7	0	c_{29}	0	c_{51}	0	c_{73}	0
c_8	−25354672	c_{30}	−32799014809633697343	c_{52}	−439261296208436545374	c_{74}	−343962399
c_9	0	c_{31}	0	c_{53}	0	c_{75}	0
c_{10}	−365670250	c_{32}	−816375902879901296921	c_{54}	2797894436524903655718	c_{76}	−166088995
c_{11}	0	c_{33}	0	c_{55}	0	c_{77}	0
c_{12}	29789795375	c_{34}	517324346869121050387	c_{56}	508678746689619550832	c_{78}	182342054
c_{13}	0	c_{35}	0	c_{57}	0	c_{79}	0
c_{14}	350400640765	c_{36}	−1041367652941099475066	c_{58}	3201411908961158733020	c_{80}	267677065
c_{15}	0	c_{37}	0	c_{59}	0	c_{81}	0
c_{16}	−17467400327971	c_{38}	3725811946796233695989	c_{60}	488666813054687153738	c_{82}	−24789787
c_{17}	0	c_{39}	0	c_{61}	0	c_{83}	0
c_{18}	−126336895666074	c_{40}	2872713371214186486271	c_{62}	−1461596764084332851792	c_{84}	1069743
c_{19}	0	c_{41}	0	c_{63}	0	c_{85}	0
c_{20}	5221986745373742	c_{42}	2019591241652839589057	c_{64}	295951128189535403620	c_{86}	3246142
c_{21}	0	c_{43}	0	c_{65}	0	c_{87}	0
c_{22}	17168504845185830	c_{44}	−871952835620434726519	c_{66}	433939135544757108160	c_{88}	−41281

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 92$, and it is the recurrence OEIS A237725 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 4$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{88} - c_1x^{88-1} - \dots - c_{88}$. Its constant term is $-c_{88} = 41281126400$, which is not zero, so $q_{\min}(0) \neq 0$ and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k\lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 88 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 88, so $\deg q = 88$, and it is monic. It holds from some index t on. From $\max(t, 4)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 88, so $q = q_{\min}$. $\square$

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 16 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 88, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry’s.

Third, the recovered recurrence was evaluated directly on the entry’s own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is -41281126400 .

Remark 4. The conjecture’s own statement was never read. The entry pins it down to the family of order-88 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

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